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Metatranscriptomics Analysis of Diatom Light Interactions and Iron Uptake

Introduction

Diatoms are unicellular phytoplankton that constitute approximately 40% of marine primary productivity and are prominent in coastal regions and open-ocean upwelling zones. Low iron bioavailability has pushed diatoms to evolve with iron-replete conditions and adapt efficient iron uptake mechanisms. Mesoscale iron fertilization experiments have established how Fe enrichments in high-nutrient, low-chlorophyll (HNLC) areas of the Southern Ocean, the Equatorial Pacific, the North Pacific and coastal upwelling zones result in large diatom blooms. Diatoms have demonstrated a unique tolerance of extremely low Fe through methods of efficient Fe uptake systems, intracellular Fe storage, or biochemical alteration of the photosynthetic Fe demand through decreased expression of the Fe-rich photosystem I (PSI) and cytochrome b6f components. Yet, comparatively less is known about how diatoms deal with episodic Fe and light availability. Iron limitation influences affects diatom and other phytoplankton productivity by decreasing the efficiency of photochemical conversion of light into chemical bond energy. The Fe additions allow for greater photosynthetic capacity and higher growth rates. In the highly dynamic coastal environment, Fe concentrations and daily surface light irradiance levels can vary by two to three orders of magnitude on short spatial and temporal scales. Diatoms that thrive in HNLC environments must have mechanisms to respond to fluxes. However, the genetic and molecular bases are unclear because few iron uptake genes have been functionally characterized and significant portions of diatom iron starvation transcriptomes are genes that encode unknown functions.

Metatranscriptomics, the study of gene expression of microbes within their natural environments, is a useful method that allows researchers to obtain whole gene expression profiling of diatom communities. Metatranscriptomics methods help to determine diversity of the active genes for iron uptake in HNLC conditions and quantifying their expression levels. This paper will examine three contemporary research studies that have used metatranscriptomics to investigate diatom (1) light-driven mechanisms and proteins for metabolism and growth (2) Fe-responsive pathways and the associated proteins and their functional roles in mediating Fe limitations. The proteins examined in these studies were death-specific proteins (DSP) in coastal diatom *Thalassiosira pseudonana*, iron concentrating proteins ISIP2a in *Phaeodactylum tricornutum*, and proteorhodopsins in oceanic diatom *Pseudo-nitzschia granii*.

Analysis

Proteorhodopsins (PR) are retinal-binding membrane proteins that function as light-driven proton pumps to generate energy. Recently, the response of eukaryotic phytoplankton to iron enrichment in iron-limited waters of the NE Pacific Ocean investigated using comparative metatranscriptome analyses (Marchetti et al., 2015). Metatranscriptomics allowed for comparison of differentially expressed genes in the iron-enriched community relative to the iron-limited community. The study discovered high expression of several eukaryotic phytoplankton-associated PR-like genes. Significantly, PR-like genes in diatoms were highly expressed under iron limitation. In iron-limited diatoms, a majority of the cellular iron is required for components of the photosynthetic electron carriers (Marchetti et al., 2015). Thus, the use of PR, which requires notably less iron than photosynthesis for ATP production, would be advantageous in iron-replete regions. This is consistent with hypotheses proposed by previous research. The presence of a PR-like gene was also discovered by sequencing the transcriptome of the oceanic diatom *Pseudo-nitzschia granii,* a diatom species that commonly dominates after iron additions to high nutrient, low chlorophyll (HNLC) regions. The *P. granii* rhodopsin gene (RHO) encodes for the seven transmembrane α-helices, a retinal binding pocket and a diagnostic residue necessary for proton transport, demonstrating that it is highly likely that the RHO is proton-pumping instead of sensory (Marchetti et al., 2015). The study further investigated how iron and light inputs influence expression patterns of *P. granii RHO* and the protein abundance. Both iron and light-limited *P. granii* experienced a reduction in specific growth rate, while photosynthetic efficiency was primarily affected by iron status. The expression of *P. granii RHO* was influenced by iron availability but not significantly affected by light availability in either iron-replete or iron-limited conditions Similar expression patterns were found at the protein level, enforcing the idea that changes in gene expression result in functional consequences. These results suggest the increase in *P. granii rhodopsin* is not a general response to growth rate reduction and is instead specific to iron-limited growth conditions. This suggests that a potential role for diatom PRs under iron limitation is to generate ATP for survival until iron becomes available for organism growth. The influence of PR on carbon cycling independent of O2-evolving mechanisms is dependent on using an alternative inorganic electron donor at the ocean surface with great abundance or fast enough turn over to influence carbon fixation.

Another type of protein studied in diatoms in low iron environments is a plastid-targeted protein in the coastal diatom *Thalassiosira pseudonana* (TpDSP1) that enhances growth during iron limitation under low light (Thamatrakoln et al. 2013). Clone lines overexpressing TpDSP1 had lower requisites for growth, greater levels of photosynthetic and carbon fixation proteins, and increased cyclic electron flow for photosystem I energy-producing pathways. Clones raised under replete conditions demonstrated reduced growth and photosynthetic rates under high light, suggesting that, although TpDSP1 offers competitive advantage in low iron conditions, cells walk an ecological tightrope boundary in regulating this protein (Thamatrakoln et al. 2013). Reverse genetics identified a death-specific protein (DSP; initially named for association with cell death) in TpDSP1 which localizes to the plastid and enhances growth during acute Fe limitation at saturated light by increasing the photosynthetic efficiency of carbon fixation. Since methods for generating knockout or silencing mutants do not currently exist in this organism, the study overexpressed TpDSP1 in its native cellular background to determine its role in the response to Fe and light availability. TpDSP1 was overexpressed in *T. pseudonana* as a C-terminal fusion with GFP. Immunoblot and flow cytometry analysis of GFP fluorescence confirmed overexpression in three independent clone lines. The results indicate that overexpression of TpDSP1 enhances growth under Fe limitation at saturating light by regulating photosynthetic electron transport but that these benefits come at a trade-off with the cost of diminished growth at high irradiance.

In the marine diatom *Phaeodactylum tricornutum,* the protein ISIP2a is utilized to concentrate Fe(III) at the cell surface as a copper-independent and thermodynamically controlled iron uptake system. ISIP2a is expressed in response to iron limitation prior to the induction of ferrireductase activity to facilitate Fe(III) uptake. ISIP2a can directly bind to Fe(III) and increase iron uptake. A knockdown of ISIP2a in *P. tricornutum* decreases iron uptake, causing weakened growth during iron limitation. To determine whether the phenotypes of the isip2a knockdown lines are caused by a change in iron uptake, this study measured the kinetics of 55Fe/59Fe uptake in the isip2a-13 and isip2a10 knockdown lines grown for varying time periods in different 6Fe/6Cu media (Morrissey et al. 2015). The findings indicate that that Fe(III) uptake was induced in iron-replete medium before induction of ferrireductase activity. In the initial response to Fe deficiency, uptake of ferric iron was more greatly affected than uptake of ferrous iron in the isip2a knockdown lines, indicating that ISIP2a may facilitate particularly Fe(III) transport. Ferric iron uptake was copper-independent, in contrast with classic ferric uptake systems in yeast and Chlamydomonas species. To determine whether ISIPs could play a role in iron uptake, genes from *P. tricornutum* were heterologously expressed in different yeast strains, including the Dfet3Dfet4 mutant, an iron-uptake-defective yeast mutant. ISIP2a, but not ISIP1 nor ISIP3, increased iron uptake. Expression of ISIP2a in both yeast and Escherichia coli enhanced the native iron uptake pathways of these species, indicating the protein binds intracellular iron. ISIP2a is hypothesized to be involved in ferric iron uptake in the initial response to iron deprivation (Morrissey et al. 2015). The impaired growth of the ISIP2a knockdown diatoms indicate that the binding and utilization of Fe(III) at the cell surface by ISIP2a is an important aspect of the initial response to iron deficiency.

Conclusion

Recently PR-like genes have been identified in some marine eukaryotic protists, including diatoms, dinoflagellates, haptophytes and cryptophytes. It has been demonstrated t in the oceanic diatom *Pseudo-nitzschia granii*, PR-like gene and protein expressions increase appreciably under iron limitation. Rhodopsin-based phototrophy may account for a proportion of energy synthesis in marine eukaryotic photoautotrophs, particularly when photosynthesis is iron-compromised. This alternative ATP-generating pathway could have significant effects on plankton community structure and global ocean carbon cycling. Additionally, in the dynamic coastal environment where cells are exposed to varying levels of Fe and light availability on short timescales, cells expressing DSP are better adapted to cope with Fe stress and have a competitive advantage. The elevated DSP levels that lead to impaired photosynthesis at saturated light forces diatoms to have trade-offs in regulation of this protein. For the protein ISIP2a, knockdown of ISIP2a in *P. tricornutum* decreases iron uptake, resulting in impaired growth and chlorosis during iron limitation. The expression of PR-like, DSP-like, and ISIP2a genes are expressed by diverse marine diatoms. Results that these genes are needed for surviving in low iron conditions indicate that it is an ecologically significant adaptation.

Literature Cited

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